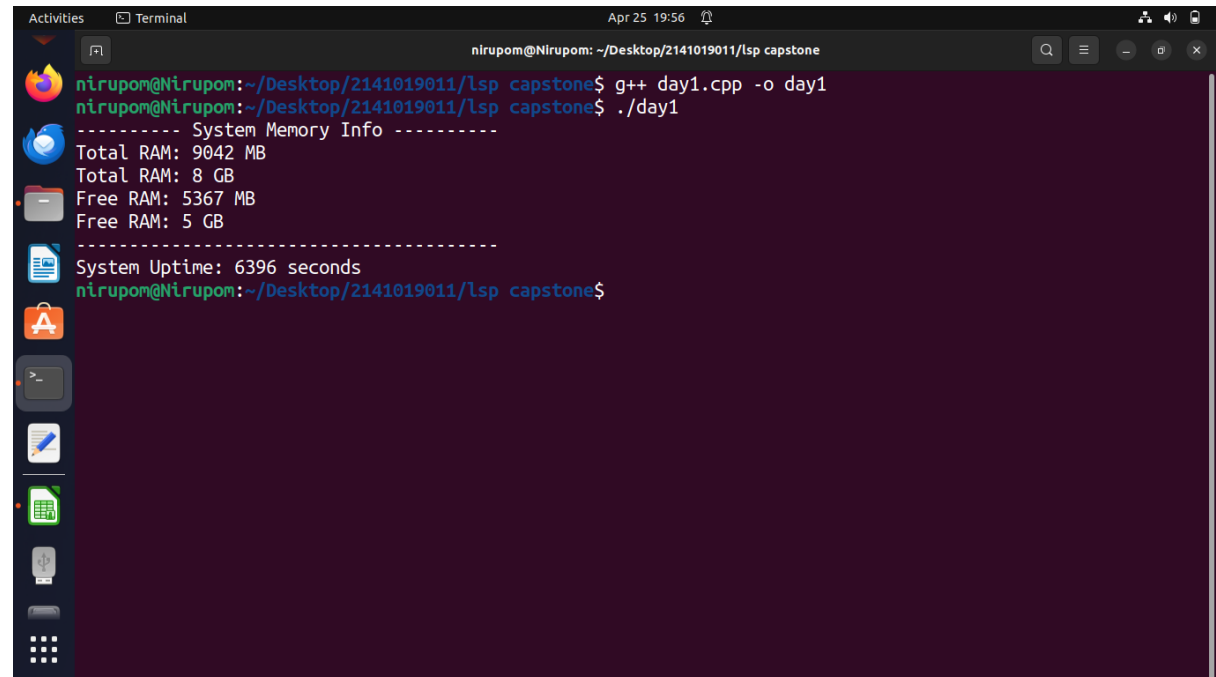


Name: Nirupom Saha

Reg. No: 2141019011

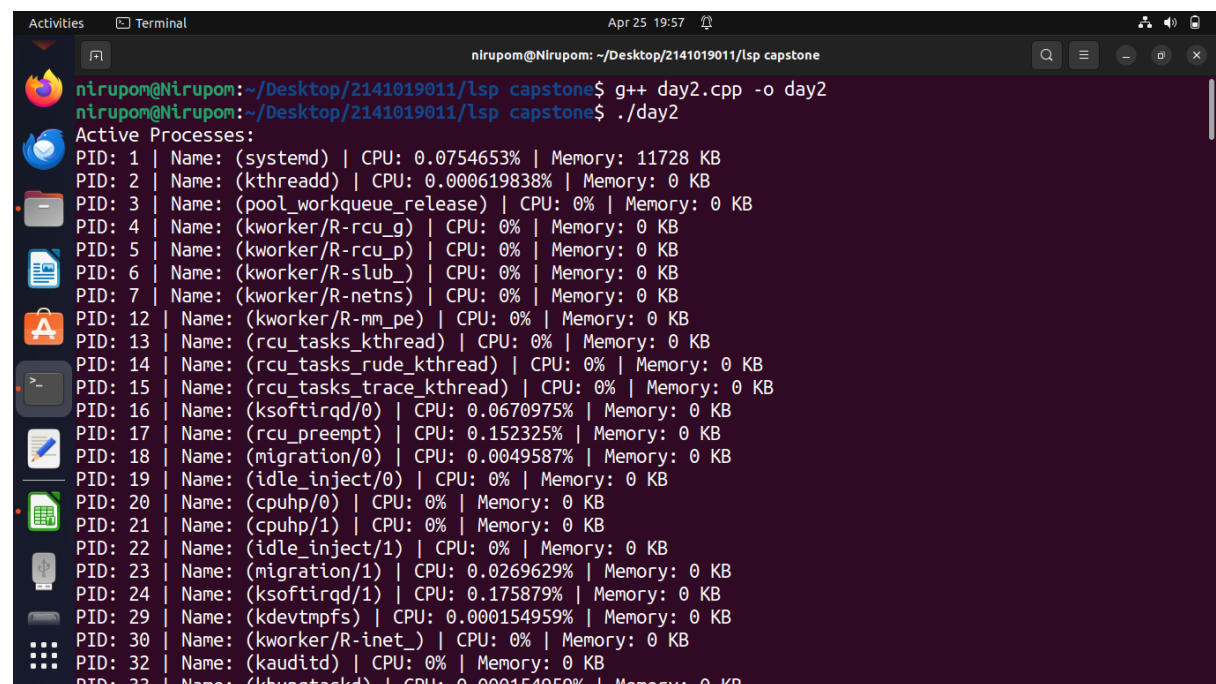
Capstone – 3

- day1.cpp



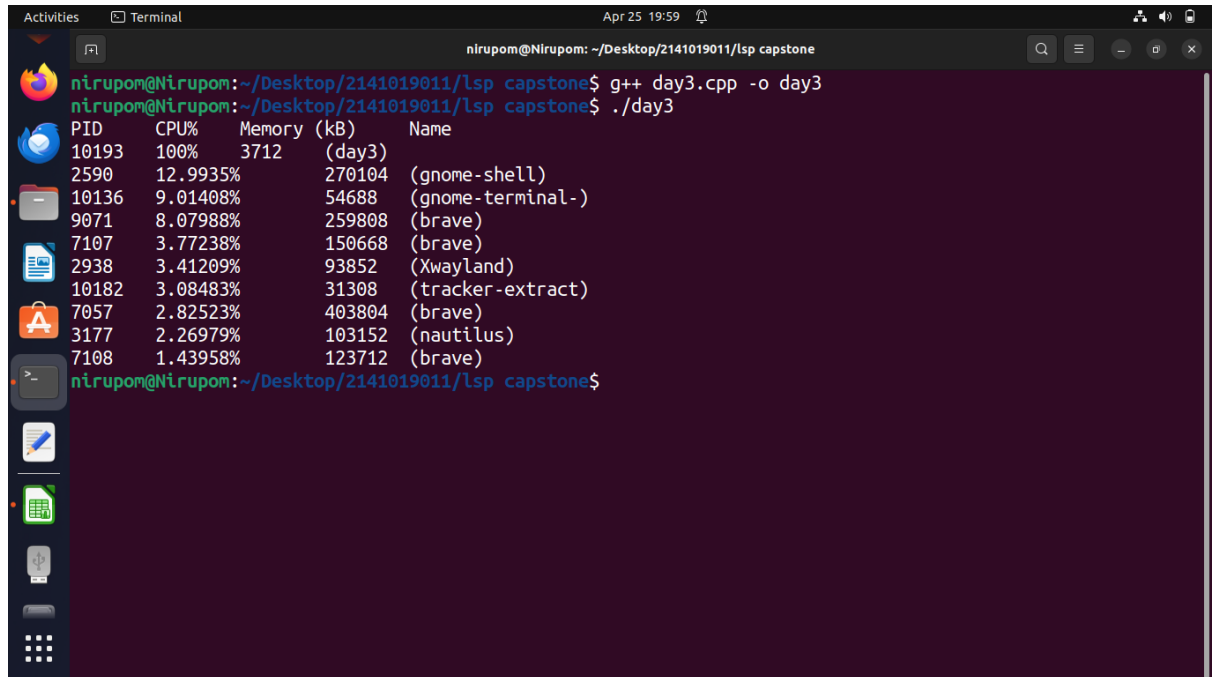
```
nirupom@Nirupom: ~/Desktop/2141019011/lsp capstone
nirupom@Nirupom:~/Desktop/2141019011/lsp capstone$ g++ day1.cpp -o day1
nirupom@Nirupom:~/Desktop/2141019011/lsp capstone$ ./day1
----- System Memory Info -----
Total RAM: 9042 MB
Total RAM: 8 GB
Free RAM: 5367 MB
Free RAM: 5 GB
-----
System Uptime: 6396 seconds
nirupom@Nirupom:~/Desktop/2141019011/lsp capstone$
```

- day2.cpp



```
nirupom@Nirupom: ~/Desktop/2141019011/lsp capstone
nirupom@Nirupom:~/Desktop/2141019011/lsp capstone$ g++ day2.cpp -o day2
nirupom@Nirupom:~/Desktop/2141019011/lsp capstone$ ./day2
Active Processes:
PID: 1 | Name: (systemd) | CPU: 0.0754653% | Memory: 11728 KB
PID: 2 | Name: (kthreadd) | CPU: 0.000619838% | Memory: 0 KB
PID: 3 | Name: (pool_workqueue_release) | CPU: 0% | Memory: 0 KB
PID: 4 | Name: (kworker/R-rcu_g) | CPU: 0% | Memory: 0 KB
PID: 5 | Name: (kworker/R-rcu_p) | CPU: 0% | Memory: 0 KB
PID: 6 | Name: (kworker/R-slub_) | CPU: 0% | Memory: 0 KB
PID: 7 | Name: (kworker/R-netns) | CPU: 0% | Memory: 0 KB
PID: 12 | Name: (kworker/R-mm_pe) | CPU: 0% | Memory: 0 KB
PID: 13 | Name: (rcu_tasks_kthread) | CPU: 0% | Memory: 0 KB
PID: 14 | Name: (rcu_tasks_rude_kthread) | CPU: 0% | Memory: 0 KB
PID: 15 | Name: (rcu_tasks_trace_kthread) | CPU: 0% | Memory: 0 KB
PID: 16 | Name: (ksoftirqd/0) | CPU: 0.0670975% | Memory: 0 KB
PID: 17 | Name: (rcu_preempt) | CPU: 0.152325% | Memory: 0 KB
PID: 18 | Name: (migration/0) | CPU: 0.0049587% | Memory: 0 KB
PID: 19 | Name: (idle_inject/0) | CPU: 0% | Memory: 0 KB
PID: 20 | Name: (cpuhp/0) | CPU: 0% | Memory: 0 KB
PID: 21 | Name: (cpuhp/1) | CPU: 0% | Memory: 0 KB
PID: 22 | Name: (idle_inject/1) | CPU: 0% | Memory: 0 KB
PID: 23 | Name: (migration/1) | CPU: 0.0269629% | Memory: 0 KB
PID: 24 | Name: (ksoftirqd/1) | CPU: 0.175879% | Memory: 0 KB
PID: 29 | Name: (kdevtmpfs) | CPU: 0.000154959% | Memory: 0 KB
PID: 30 | Name: (kworker/R-inet_) | CPU: 0% | Memory: 0 KB
PID: 32 | Name: (kauditd) | CPU: 0% | Memory: 0 KB
PID: 33 | Name: (khungtaskd) | CPU: 0.000154959% | Memory: 0 KB
```

- **day3.cpp**

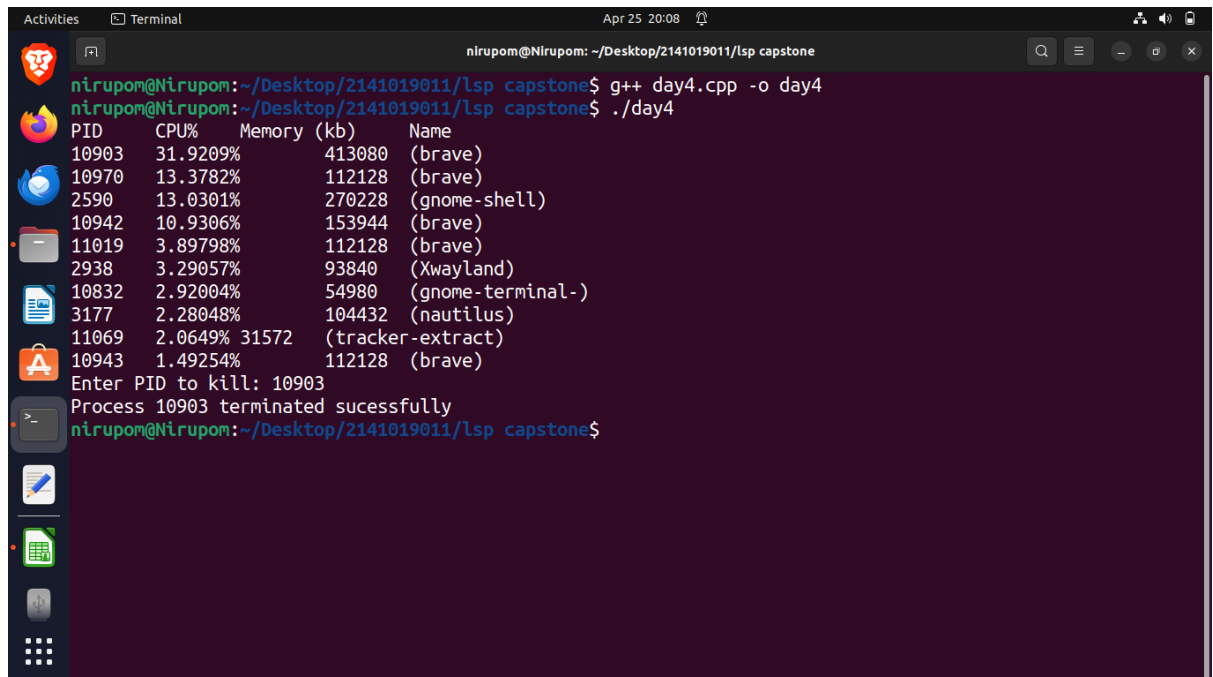


```
nirupom@Nirupom: ~/Desktop/2141019011/lsp capstone$ g++ day3.cpp -o day3
nirupom@Nirupom: ~/Desktop/2141019011/lsp capstone$ ./day3
```

| PID   | CPU%     | Memory (kB) | Name              |
|-------|----------|-------------|-------------------|
| 10193 | 100%     | 3712        | (day3)            |
| 2590  | 12.9935% | 270104      | (gnome-shell)     |
| 10136 | 9.01408% | 54688       | (gnome-terminal-) |
| 9071  | 8.07988% | 259808      | (brave)           |
| 7107  | 3.77238% | 150668      | (brave)           |
| 2938  | 3.41209% | 93852       | (Xwayland)        |
| 10182 | 3.08483% | 31308       | (tracker-extract) |
| 7057  | 2.82523% | 403804      | (brave)           |
| 3177  | 2.26979% | 103152      | (nautilus)        |
| 7108  | 1.43958% | 123712      | (brave)           |

```
nirupom@Nirupom: ~/Desktop/2141019011/lsp capstone$
```

- **day4.cpp**



```
nirupom@Nirupom: ~/Desktop/2141019011/lsp capstone$ g++ day4.cpp -o day4
nirupom@Nirupom: ~/Desktop/2141019011/lsp capstone$ ./day4
```

| PID   | CPU%     | Memory (kb) | Name              |
|-------|----------|-------------|-------------------|
| 10903 | 31.9209% | 413080      | (brave)           |
| 10970 | 13.3782% | 112128      | (brave)           |
| 2590  | 13.0301% | 270228      | (gnome-shell)     |
| 10942 | 10.9306% | 153944      | (brave)           |
| 11019 | 3.89798% | 112128      | (brave)           |
| 2938  | 3.29057% | 93840       | (Xwayland)        |
| 10832 | 2.92004% | 54980       | (gnome-terminal-) |
| 3177  | 2.28048% | 104432      | (nautilus)        |
| 11069 | 2.0649%  | 31572       | (tracker-extract) |
| 10943 | 1.49254% | 112128      | (brave)           |

```
Enter PID to kill: 10903
Process 10903 terminated successfully
nirupom@Nirupom: ~/Desktop/2141019011/lsp capstone$
```

- **day5.cpp**

Terminal window showing process list and kill command execution. The process list includes PID, CPU%, Memory (kb), and Name. The process (brave) with PID 11484 is highlighted. The kill command is entered, and the process is terminated successfully.

| PID   | CPU%     | Memory (kb) | Name              |
|-------|----------|-------------|-------------------|
| 11484 | 17.3246% | 412484      | (brave)           |
| 2590  | 13.042%  | 270428      | (gnome-shell)     |
| 11558 | 7.18606% | 113152      | (brave)           |
| 11524 | 4.06331% | 157052      | (brave)           |
| 11737 | 3.23326% | 31096       | (tracker-extract) |
| 2938  | 3.22701% | 93892       | (Xwayland)        |
| 11742 | 3.0303%  | 3328        | (day5)            |
| 11621 | 2.33077% | 113152      | (brave)           |
| 3177  | 2.28294% | 105912      | (nautilus)        |
| 10832 | 1.85613% | 55148       | (gnome-terminal-) |

Enter PID to kill: 11484  
Process 11484 terminated successfully  
Press 'q' to quit or Enter to refresh:

Terminal window showing the updated process list after killing PID 11484. The process list includes PID, CPU%, Memory (kb), and Name. The process (brave) with PID 11484 is no longer present.

| PID   | CPU%      | Memory (kb) | Name              |
|-------|-----------|-------------|-------------------|
| 2590  | 13.0293%  | 274348      | (gnome-shell)     |
| 11779 | 8.125%    | 31468       | (tracker-extract) |
| 2938  | 3.21041%  | 86420       | (Xwayland)        |
| 3177  | 2.30006%  | 110788      | (nautilus)        |
| 10832 | 1.69667%  | 55148       | (gnome-terminal-) |
| 11762 | 1.10619%  | 3456        | (systemd-hostnam) |
| 5800  | 1.02981%  | 289056      | (soffice.bin)     |
| 3111  | 0.702577% | 3100        | (VBoxClient)      |
| 2786  | 0.401019% | 12212       | (ibus-daemon)     |
| 2968  | 0.27912%  | 42928       | (tracker-miner-f) |

Enter PID to kill: